CHRISTIAN OMAR PAYÁN TORRÓNTEGUI

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SUMMARY

Robotics & Digital Systems Engineering student focused on product design, advanced manufacturing, and robotics. Proven experience in 3D modeling, additive manufacturing, and Python simulation. Seeking to leverage analytical design to bridge creative design with technical feasibility within Railway Engineering.

EDUCATION

B.S. Robotics and Digital Systems Graduating Expected December 2026
Tecnológico de Monterrey
Zapopan, Jalisco
Relevant coursework: Python Programming, CAD Design, Advanced 3D modeling

TECHNICAL SKILLS

Programming & Languages: Python, C++, Spanish (Native), English (B2)
Robotics & Simulation: ROS 2, Gazebo, URDF
Design & Electronics: SOLIDWORKS, Fusion 360, Eagle
Tools & Methodologies: Git, Agile, Scrum, Advanced FDM 3D Printing, Linux (Ubuntu), Docker

PROFESSIONAL EXPERIENCE

Industrial Mounts & Robotic Support Interfaces, Zapopan, Jalisco: Mechanical & Product Designer Feb-Jun 2026

- Engineered mechanical interfaces for VR controllers (Meta Quest 3), developing a 3-DOF system that reduced fabrication costs by 95% vs commercial alternatives
- Designed and prototyped a custom mini PC enclosure for robust deployment in industrial environments, utilizing 3D-printed fiberglass-reinforced PETG for its ideal thermal and mechanical properties

ACADEMIC & TECHNICAL PROJECTS

Open-Source Gastrointestinal Medical Phantom Ago-Dic 2025 – Feb-Jun 2026
Medical phantom via advanced FDM 3D printing

- Engineered phantom via advanced FDM 3D printing (TPU/PETG), reducing prototyping costs by 80% vs commercial alternatives
- Managed development using Agile methodologies to iterate rapidly on anatomical accuracy and material constraints

Port Automation & Drone Sim Ago-Dic 2025
Developed a port simulation environment utilizing ROS 2 and Gazebo in partnership with OMRON

- Engineered URDF models and C++/Python ROS 2 nodes for kinematic control and reliable flight telemetry
- Designed & 3D-printed custom hardware: drone mirror/lens mounts, and an AMR bracket (LiDAR, ArUco, Jetson)

DigiMix Ago-Dic 2024
Designed an embedded system integrating custom hardware and digital interfaces with Intel

- Designed custom PCB and 3D-printed casing; engineered hardware using dual-core STM32 and ESP32 (WebSockets)
- Applied Scrum to manage lifecycle, integrating hardware and software architectures with zero logic faults